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- All problems are worth an equal weight, even if some may be harder than others.
 - Answers may be typed or handwritten, either on paper or a note taking device. There is no need to copy the problem statements, but please upload a single pdf file.
 - Late submissions will be worth significantly less points.
 - Circle your final answer, but make sure to include the work to get there.
 - It is OK to work with others, as long as you acknowledge them in your answers and write down your solutions in your own words. It is not OK to copy a solution from whatever source.
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1. A certain insect population in a forest is becoming extinct according to the equation $P'(t) = -\lambda P(t)$ (where $P(t)$ represents the population after t months and $\lambda > 0$ is a parameter that needs to be determined).

- a) If the population decreased to 60% its original size in 3 months, what is the value of λ ?
- b) Solve the differential equation using the separation of variables method.
- c) If the initial insect population was estimated at 1,000,000, how much will it be after 6 months have passed since then?
- d) Does the model predict that the population will ever become extinct?

2. Answer the following items about mixing problems:

- a) A giant pot contains 5L of water, initially without any salt. Suppose that there is water containing 2g/L of salt flowing into the pot at all times at a rate of 0.5L/min, with that salt instantly spreading homogeneously in the water inside. Suppose also that the contents of the pot are leaking at the same rate of 0.5L/min. What will be the **concentration** of salt inside after 20 min?
- b) Suppose instead that the water flows in at a rate of 1L/min, everything else being equal. Write down the IVP that is satisfied by $m(t)$, the mass of salt inside the pot at time t (but do not solve). *Hint: What is the formula of the volume of water in the pot at time t ? Use it to figure out how much salt is leaving the pot every minute.*

3. An egg initially at 25°C is put in boiling water (at 100°C). Suppose the temperature of the egg follows Newton's law of cooling (should it be called heating?) and that the water remains at 100°C at all times (boiling water actually does that). If it takes 4 minutes for the egg to reach 60°C , how much time since the start will it take to reach 71°C for optimum hard-boiled deliciousness?

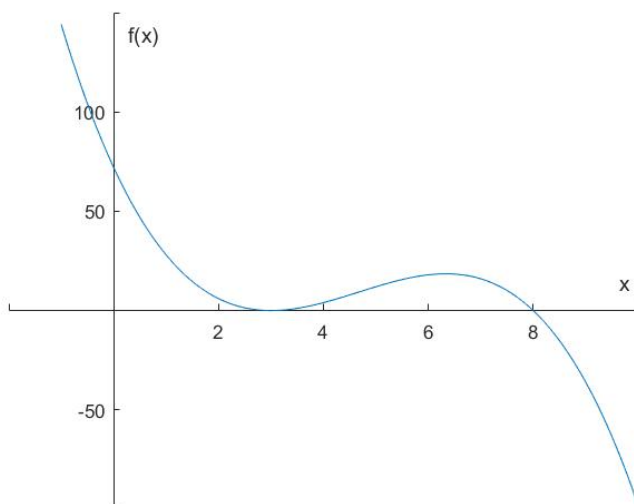
4. Find the general solution of these differential equations using separation of variables:

- a) $y' = \frac{3y - 2}{3t}$
- b) $y' = y^2 - 9$
- c) $y' = \frac{t^2 + 1}{y - 1}$
- d) $y' = \frac{1}{(t^2 + 2)(y - 1)}$

5. Sketch the slope field and the graphs of some sample solutions of the differential equation below:

$$y' = (t - 1)(2y + t)$$

6. Let f be the function whose graph is depicted below.



a) Sketch the slope field of the equation $y' = f(y)$.

b) Sketch the slope field of the equation $y' = f(t)$.

7. Use the Euler method with step size $\Delta t = 0.2$ to find an approximation for $y(2)$ in the following IVP:

$$\begin{cases} y' = \sqrt{t + y} \\ y(1) = 1 \end{cases}$$

You will need a calculator for this. Keep numbers approximated to 3 decimal places, rounding off when needed.

8. Suppose we wanted to use the Euler method to successfully approximate $T(1)$ in the IVP

$$\begin{cases} T' = 5(100 - T) \\ T(0) = 200 \end{cases}$$

to a precision of 0.05. How small of a Δt will guarantee such precision? *Hint: We know the formula of the solution function. Use that to determine a bound for $|T''|$.*

9. Consider the equation

$$y' = -y^2 + y + 2yt^2 + 2t - t^2 - t^4$$

a) Verify that both $u = t^2$ and $v = t^2 + 1$ are solutions.

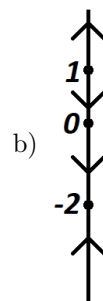
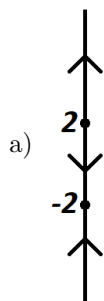
b) Given the item above, what does the existence and uniqueness theorem imply about the solution $y(t)$ that satisfies $y(3) = 19/2$? Remember to verify that the theorem can be applied.

10. Bob initially has \$1,000 in his bank account, which earns interest of 2% per year compounded continuously. He also deposits \$100 every year into the account.

- Write an IVP for $S(t)$, the money in his account at time t measured in years.
- Find the general solution to the differential equation three times, by three different methods: separation of variables, undetermined coefficients, and an integrating factor. Then find the value of C in the general solution using the initial condition.
- How much time will it take for the account to reach \$13,000?

Remark: Interest in real life is not actually compounded on a continuous basis, like your differential equation suggests, but rather once per month. The deposits in this problem are also made once per year, not continuously. But this turns out to be an accurate way of modeling the problem, and the formula you obtain can be used for easily computing earnings with interest.

11. For each of the phase lines below, come up with a function f that produces them in the differential equation $y' = f(y)$.



12. Can the derivative criterion be used to classify the equilibrium points of the equation below as sinks or sources? If not, classify them by plugging-in points and drawing the phase line.

$$y' = y^3 \ln \left(\frac{y^2 + 5}{9} \right)$$

13. Let $y' = (y - 1)(y - 3)$.

- Sketch the phase line, then based on it sketch the graphs of sample solutions.
- Solve the differential equation with $y(0) = 2$ and use calculus to sketch the graph of the solution. Compare to the corresponding graph from item (a).

14. Draw the bifurcation diagram of the following equations with respect to the parameter A , and find bifurcation values if any exist.

- $y' = (y + A)^2 - 25$.
- $y' = \cos y + A$.

15. Consider the equation

$$y' = y \cos t + \frac{1}{5}(1 - t \cos t)$$

- Verify that $y = t/5$ is a solution.
- Use the Extended Linearity Principle to find the general solution.